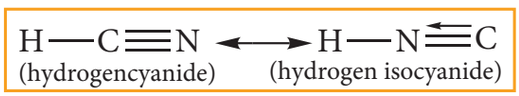
(e) Tautomerism: It is a special type of functional isomerism in which a single compound exists in two readily inter convertible structures that differ markedly in the relative position of atleast one atomic nucleus, generally hydrogen. The two different structures are known as tautomers. There are several types of tautomerism and the two important types are dyad and triad systems.

(i) Dyad system: In this system hydrogen atom oscillates between two directly linked polyvalent atoms. Eg:



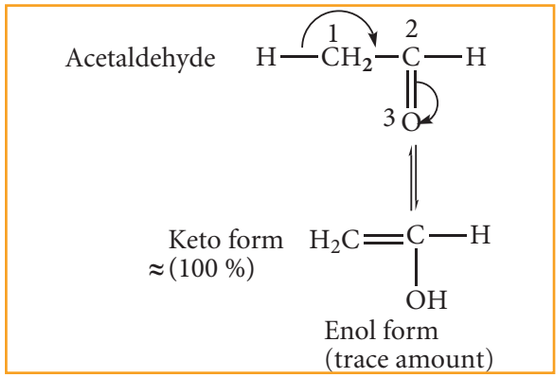
In this example hydrogen atom oscillates between carbon & nitrogen atom

Evaluate Yourself

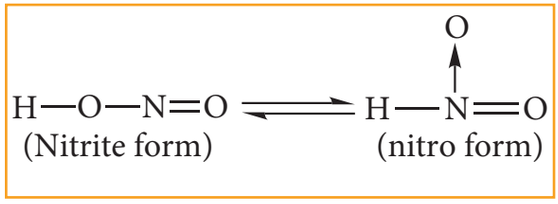
4) Write all the possible isomers of molecular formula $C_4H_{10}O$ and identify the isomerisms found in them.

(ii) Triad system: In this system hydrogen atom oscillates between three polyvalent atoms. It involves 1,3 migration of hydrogen atom from one polyvalent atom to other within the molecule. The most important type of triad system is keto–enol tautomerism and the two groups of tautomers are ketoform and enol-form. The polyvalent atoms involved are one oxygen and two carbon atoms. Enolisation is a process in which keto-form is converted to enol form. Both tautomeric forms are not equally stable. The less stable form is known as labile form

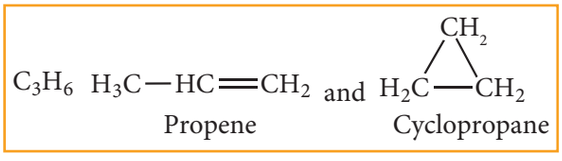
Example:

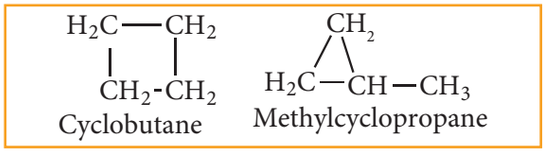


Nitro-aci tautomerism.



(f) Ring chain isomerism: In this type of isomerism, compounds having same molecular formula but differ in terms of bonding of carbon atom to form open chain and cyclic structures for eg:





11.5.2 Stereoisomerism:

The isomers which have same bond connectivity but different arrangement of groups or atoms in space are known as stereoisomers. This branch of chemistry dealing with the study of three-dimensional nature (spactial arrangement) of molecules is known as stereo chemistry. The metabolic activities in living organisms, natural synthesis and drug synthesis involve various stereoisomers.

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